POWER BI REPORT

WEEK 5

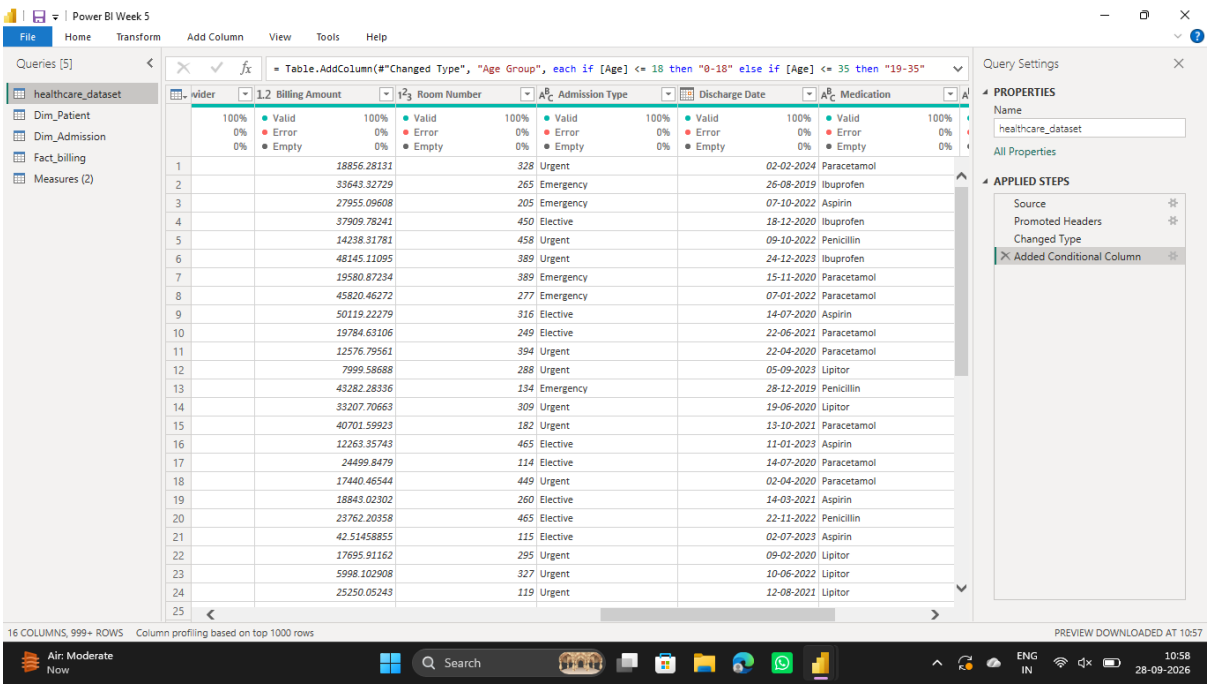
Healthcare Data Cleaning & Data Modeling

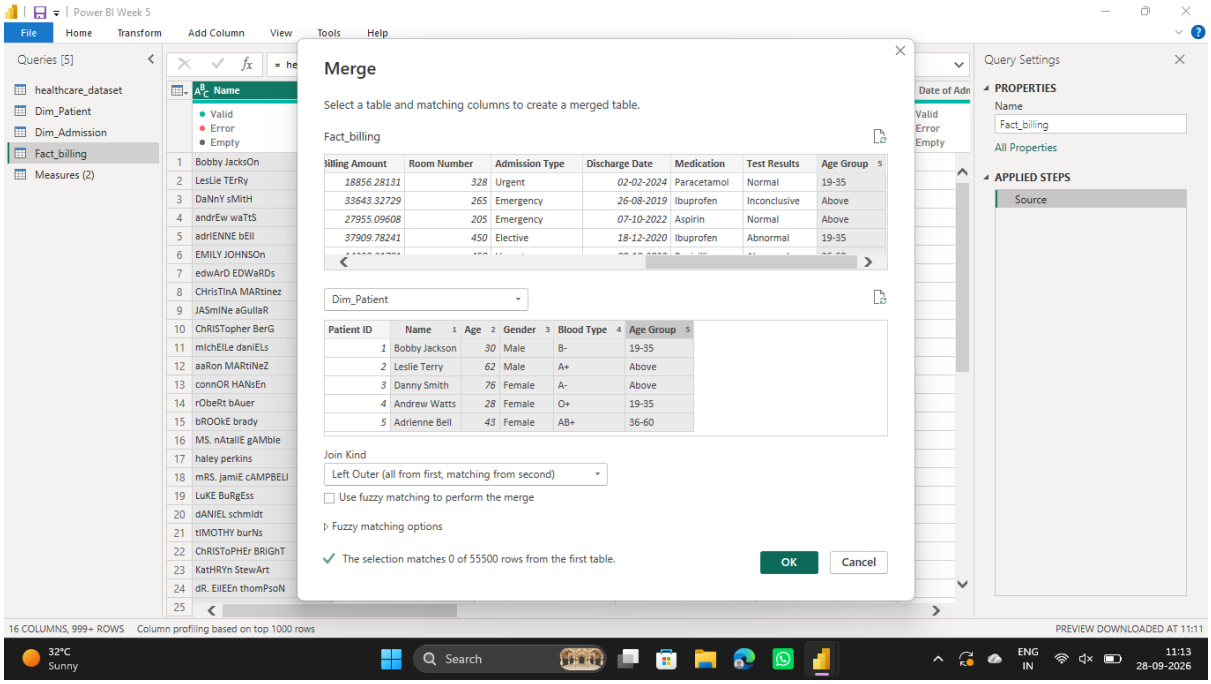
Dataset: Healthcare Dataset

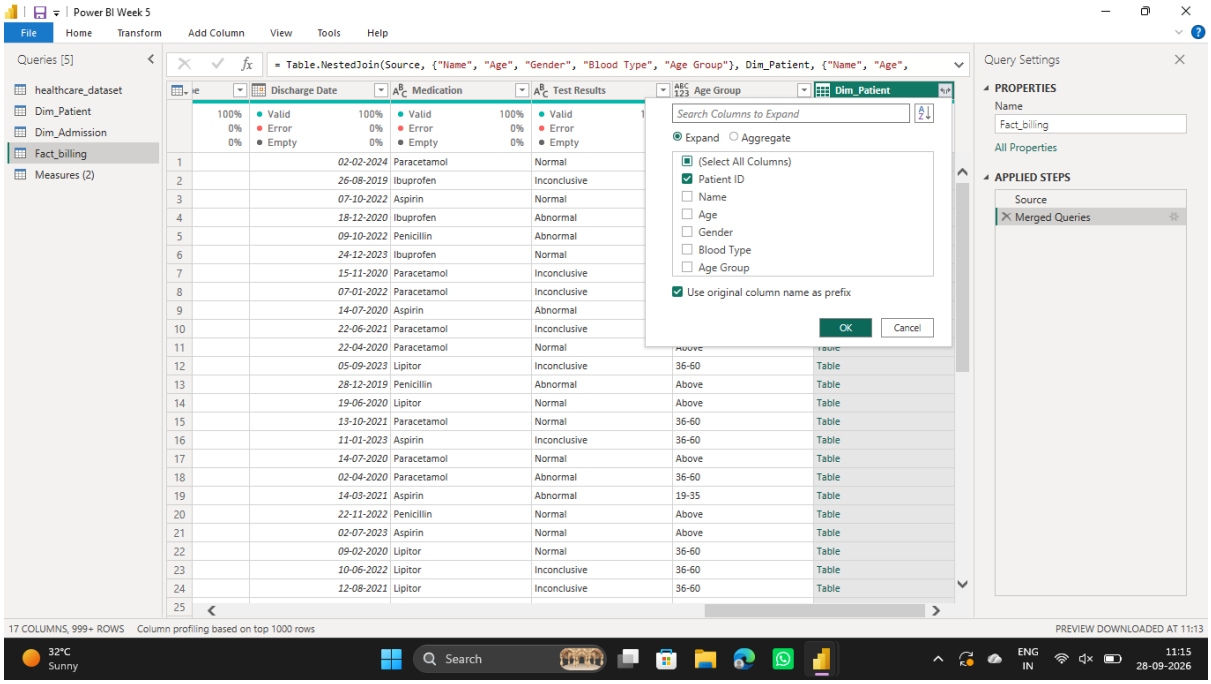
Problem Statement:
The healthcare dataset contains patient, admission, billing and insurance-related information. Before creating meaningful visualizations, the data needs to be transformed and organized into separate tables. Power Query and Power BI modelling features are used to prepare the dataset for efficient analysis.

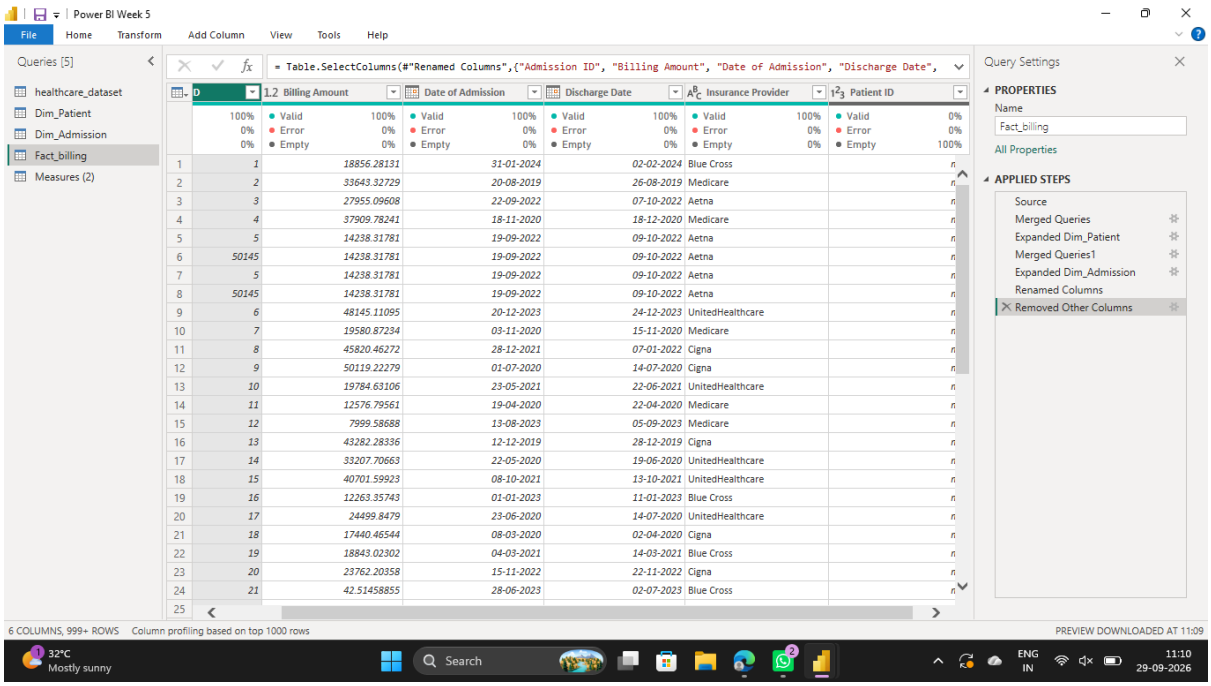
Objectives

- Prepare the healthcare dataset for analysis.
- Transform and organize important columns.
- Create separate tables for different data categories.
- Establish relationships between the tables.
- Remove unnecessary fields and improve the data model.
- Prepare the final model for dashboard development.




Power BI Week 5
File
Home
Transform
Add Column
View
Tools
Help
Queries [5]
healthcare_dataset
Dim_Patient
Dim_Admission
Fact_billing
Measures (2)
Merge
Select a table and matching columns to create a merged table.
Fact_billing
Billing Amount
Room Number
Admission Type
Discharge Date
Medication
Test Results
Age Group
18856.28131
328
Urgent
02-02-2024
Paracetamol
Normal
19-35
33643.32729
265
Emergency
26-08-2019
Ibuprofen
Inconclusive
Above
27955.09608
205
Emergency
07-10-2022
Aspirin
Normal
Above
37909.78241
450
Elective
18-12-2020
Ibuprofen
Abnormal
19-35
Dim_Patient
Patient ID
Name
Age
Gender
Blood Type
Age Group
1
Bobby Jackson
30
Male
B-
19-35
2
Leslie Terry
62
Male
A+
Above
3
Danny Smith
76
Female
A-
Above
4
Andrew Watts
28
Female
O+
19-35
5
Adrienne Bell
43
Female
AB+
36-60
Join Kind
Left Outer (all from first, matching from second)
Use fuzzy matching to perform the merge
Fuzzy matching options
The selection matches 0 of 55500 rows from the first table.
OK
Cancel
Query Settings
PROPERTIES
Name
Fact_billing
All Properties
APPLIED STEPS
Source
16 COLUMNS, 999+ ROWS
Column profiling based on top 1000 rows
PREVIEW DOWNLOADED AT 11:11
32°C
Sunny
Search
ENG
IN
11:13
28-09-2026


Power BI Week 5
File
Home
Transform
Add Column
View
Tools
Help
Queries [5]
healthcare_dataset
Dim_Patient
Dim_Admission
Fact_billing
Measures (2)
= Table.NestedJoin(Source, {"Name", "Age", "Gender", "Blood Type", "Age Group"}, Dim_Patient, {"Name", "Age",
Discharge Date
Medication
Test Results
Age Group
Dim_Patient
Search Columns to Expand
Expand
Aggregate
(Select All Columns)
Patient ID
Name
Age
Gender
Blood Type
Age Group
Use original column name as prefix
OK
Cancel
02-02-2024 Paracetamol Normal
26-08-2019 Ibuprofen Inconclusive
07-10-2022 Aspirin Normal
18-12-2020 Ibuprofen Abnormal
09-10-2022 Penicillin Abnormal
24-12-2023 Ibuprofen Normal
15-11-2020 Paracetamol Inconclusive
07-01-2022 Paracetamol Inconclusive
14-07-2020 Aspirin Abnormal
22-06-2021 Paracetamol Inconclusive
22-04-2020 Paracetamol Normal
05-09-2023 Lipitor Inconclusive 36-60 Table
28-12-2019 Penicillin Abnormal Above Table
19-06-2020 Lipitor Normal Above Table
13-10-2021 Paracetamol Normal 36-60 Table
11-01-2023 Aspirin Inconclusive 36-60 Table
14-07-2020 Paracetamol Normal Above Table
02-04-2020 Paracetamol Abnormal 36-60 Table
14-03-2021 Aspirin Abnormal 19-35 Table
22-11-2022 Penicillin Normal Above Table
02-07-2023 Aspirin Normal Above Table
09-02-2020 Lipitor Normal 36-60 Table
10-06-2022 Lipitor Inconclusive 36-60 Table
12-08-2021 Lipitor Inconclusive 36-60 Table
Query Settings
PROPERTIES
Name
Fact_billing
All Properties
APPLIED STEPS
Source
Merged Queries
17 COLUMNS, 999+ ROWS
Column profiling based on top 1000 rows
PREVIEW DOWNLOADED AT 11:13
32°C
Sunny
Search
ENG
IN
11:15
28-09-2026


Power BI Week 5
File
Home
Transform
Add Column
View
Tools
Help
Queries [5]
healthcare_dataset
Dim_Patient
Dim_Admission
Fact_billing
Measures (2)
= Table.SelectColumns(#"Renamed Columns",{"Admission ID", "Billing Amount", "Date of Admission", "Discharge Date",
ID
Billing Amount
Date of Admission
Discharge Date
Insurance Provider
Patient ID
Query Settings
PROPERTIES
Name
Fact_billing
All Properties
APPLIED STEPS
Source
Merged Queries
Expanded Dim_Patient
Merged Queries1
Expanded Dim_Admission
Renamed Columns
Removed Other Columns
6 COLUMNS, 999+ ROWS
Column profiling based on top 1000 rows
PREVIEW DOWNLOADED AT 11:09
32°C
Mostly sunny
Search
ENG
IN
11:10
29-09-2026


Dim_Patient
Age
Age Group
Blood Type
Gender
Name
Patient ID
Dim_Admission
Admission ID
Admission Type
Doctor
Hospital
Medical Condition
Medication
Room Number
Test Results
Fact_billing
Admission ID
Admission Type
Age
Age Group
Billing Amount
Blood Type
1
*

Conclusion

During Week 5, the healthcare dataset was prepared and organized in Power BI for further analysis. The data was transformed by cleaning important fields, creating required tables, and connecting related information through suitable relationships. The final data model provides a clear structure for patient, admission, and billing information. These transformations make the dataset easier to manage and provide a strong foundation for creating meaningful dashboards and performing further analysis in the upcoming stages.